

Special Topics Course in Computational Science
CMSC6903: MPI Programming for High Performance Computing
Fall Semester 2012-2013

Motivation

This is a hands-on 2-credit course on the construction (design and development) of MPI-based parallel software using C (a basic introduction to C-language will be provided). The course serves a dual purpose.

1. It is a lab portion of CMSC6930. As such, it should be taken by those that will take CMSC6930.
2. It can be considered as a self-contained course. As such, it can be taken by anybody interested in developing the ability to program in distributed computing environments.

Upon completion of this course, students will be able to:

- Program parallel architectures in C (MPI programming with other languages is straightforward)
- Debug and optimize parallel programs on High Performance Computers at ACEnet.

Course evaluation

Assignments 25%. Spontaneous quizzes 25%. Project 50%.

Lecturer

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Textbooks

- Class notes.
- Using MPI: Portable Parallel Programming with the Message Passing Interface, 2nd ed. W. Gropp, E. Lusk and A. Skjellum. MIT Press.
- Parallel programming with MPI. Peter Pacheco. Morgan Kaufman.
- The C Programming Language. B. Kernigham and D. Ritchie. Prentice Hall.

Pre-requisite

Permission of the Computational Sciences Program Director. Having completed previous programming course(s).

Outline

1. Introduction to the C-language (2 weeks)
2. Introduction to distributed memory hardware and software (1 week)
3. Message Passing Interface (MPI) (7 weeks)
 - a. Point to point communications
 - b. Collective communication
 - c. Grouping data for communication
 - d. Dealing with I/O
 - e. Algorithms in MPI
4. MPI Software Performance Analysis (2 weeks)
5. Parallel libraries (1 week)
6. No final exam. Only spontaneous quizzes, assignments and project used as evaluation.